

University of Natural Resources and Life Sciences, Vienna (BOKU)

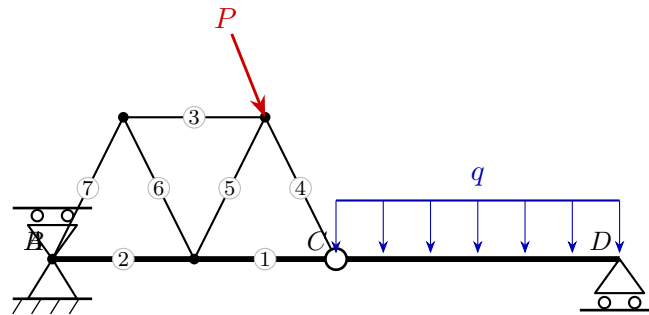
VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

Date: 26.06.2026

Given system

The planar system consists of bending beams and a truss connected by hinges.



Given quantities: $a = 3\text{ m}$ $\ell = 1,5\text{ m}$ $h = 1,5\text{ m}$ $q = 4\text{ kN/m}$ $P = 20\text{ kN}$

Tasks

1. Determine the degree of static determinacy.
2. Compute all support reactions and hinge forces.
3. Determine all truss member forces (tension/compression).
4. Draw the section forces N , V , M of the bending beams.